mycorrhizal association. Two types of branches called long shoot and dwarf shoot are present. Stem shows secondary growth. Spores are produced in cones. Pollen tube helps in fertilization. The endosperm is haploid. Alternation of generation is present

Cycas belongs to cycadopsida. The plant body is sporophyte and looks like a small palm tree. Apart from taproot coralloid roots are present. It is dioecious, microsporophylls are organized into male cone. Ovules are borne on megasporophylls which are not organized into cone. Fertilization results in zygote and it develops into embryo. Alternation of generation is present.

Angiosperms are highly evolved plant group and their ovules remain enclosed in an ovary. A wide range of habit is present. These include trees, shrubs, herbs, climbers, lianas. Double fertilization is present. The endosperm is triploid. They are classified into dicotyledons and monocotyledons.

Evaluation



1. Which of the plant group has gametophyte as a dominant phase?
   a. Pteridophytes
   b. Bryophytes
   c. Gymnosperms
   d. Angiosperms
2. Which of following represents gametophytic generation in pteridophytes?
   a. Prothallus
   b. Thallus
   c. Cone
   d. Rhizophore
3. The haploid number of chromosome for an angiosperm is 14, the number of chromosome in its endosperm would be
   a. 7 b. 14 c. 42 d. 28
4. In gymnosperm endosperm is formed
   a. At the time of fertilization
   b. Before fertilization
   c. After fertilization
   d. Along with the development of embryo
5. Differentiate halpontic and diplontic life cycle.
6. What is plectostele? give example.
7. What do you infer from the term pycnoxylic?
8. Mention two characters shared by gymnosperms and angiosperms.
9. Do you think shape of chloroplast is unique for algae. Justify your answer?
10. Do you agree with the statement 'Bryophytes need water for fertilization'? Justify your answer.
11. List out the classes of algae.
12. Mention the pigments and storage food of Rhodophyceae.
13. What is Nucule?
14. Differentiate nodal and internodal cells of *Chara*.
15. Explain the internal structure of *Cycas* rachis.

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